Curriculum Vitae

Mao Nagamine

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Research Interests

- A-hypergeometric (GKZ) systems; logarithmic series solutions and Frobenius' method.
- Toric ideals, Gröbner bases, standard pairs, and regular triangulations.
- Combinatorial commutative algebra; Stanley–Reisner rings and Hilbert series.

Education

Ph.D. program in Mathematics, Hokkaido University, April 2024–present. Graduate School of Science. Doctoral advisors: Mutsumi Saito (through second year); Masanori Asakura (third year).

M.Sc. in Science (Mathematics), Hokkaido University, March 2024. Graduate School of Science. Advisor: Mutsumi Saito.

B.Sc. in Science (Mathematics), Hokkaido University, March 2022.

Undergraduate studies in Mathematical Sciences, Kwansei Gakuin University, April 2018–March 2020.

Appointments

Research Staff, Hokkaido University, July 2024–present.

Research Assistant, Hokkaido University, Mathematical and Data Science Education and Research Center, April 2024–present.

Part-time Lecturer, Hokkaido University of Education, Hakodate Campus, October 2025–September 2026.

Research Assistant, Hokkaido University, April 2023–March 2024.

Awards and Fellowships

- Outstanding Research Encouragement Award, Graduate School of Science, Hokkaido University (April 2025).
- EXEX Doctoral Fellowship, DX Doctoral Talent Program, Hokkaido University (April 2024–present).
- Museum Meister Certificate, Hokkaido University Museum (May 2024).

Research Funding

Research on A-hypergeometric differential equation systems, JST SPRING doctoral research support, April 2024–March 2027.

Publications and Preprints

1. M. Nagamine, "A-hypergeometric Series with Parameters in the Core," *Mathematical Journal of Okayama University*, accepted October 16, 2025 (forthcoming). arXiv:2404.11085.
2. M. Nagamine, "Hilbert Series and Logarithmic Degrees of A-Hypergeometric Series," preprint (2026), under review. arXiv:2608.01778.
3. M. Nagamine, "Core Coordinates and Negative Support in A-Hypergeometric Systems," preprint (2026), submitted. arXiv:2610.04369.

Other Writing

M. Nagamine, "A-hypergeometric series with parameters in the core," *Hokkaido University Technical Report Series in Mathematics*, vol. 189, pp. 197–204 (2025), in Japanese.

Talks and Presentations

- "Conditions for applying Frobenius' method to A-hypergeometric systems," Representation Theory Seminar, Hokkaido University (February 20, 2026). [Title translated from Japanese.]
- "A-hypergeometric series with parameters in the core," *Arithmetic of Hypergeometric Functions and Related Topics*, Hokkaido University (July 15, 2025). **Invited talk**.
- "A-hypergeometric series with parameters in the core," 29th Young Algebraists' Seminar, Osaka University (March 14, 2025). [Title translated from Japanese.]
- "A-hypergeometric series with parameters in the core," 21st Mathematics Comprehensive Young Researchers Meeting (MCYR), Hokkaido University (March 7, 2025). [Title translated from Japanese.]
- "A-hypergeometric series with parameters in the core," 8th Newcomers' Seminar in Mathematical Sciences, Nagoya University (February 12, 2025). [Title translated from Japanese.]
- "A-hypergeometric series with parameters in the core," Autumn Meeting of the Mathematical Society of Japan, Osaka University (September 6, 2024).
- "Construction of the logarithmic solutions of unimodular A-hypergeometric systems," 30th Hokkaido Special Functions Seminar, Hokkaido University (January 30, 2024).

Teaching Experience

Hokkaido University of Education, Hakodate Campus: part-time lecturer (October 2025–September 2026).

Courses taught include Analysis I (infinite series), 2026.

Academic Service

- Organizing committee, 20th–22nd Mathematics Comprehensive Young Researchers Meetings (MCYR), 2023–2026.
- Judge, Hokkaido University Undergraduate Thesis Poster Presentations, 2025 and 2026; event staff, 2024.
- Planning and organization of a graduate-student research exhibition, NoMaps 2024, Sapporo (September 2024).

Outreach

- Science communication and educational events at Hokkaido University, including the Hokkaido University Museum and graduate-student outreach activities (2023–2024).

Professional Membership

Mathematical Society of Japan.